

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Software engineer specializing in LLM and agentic systems -- building multi-model AI pipelines (Claude, GPT-4o, Gemini) and MCP tooling in production at Isomer AI, with daily Claude Code use across the full development lifecycle. Delivered 394 commits and 7 production deployments across 10+ backend features in a fast-moving startup environment. Built agentic orchestration (Temporal.io + Kafka, 2,000 req/sec), compliance-ready RAG for regulated domains, and Kafka-driven data pipelines -- comfortable taking systems from 0 to 1 and rethinking them mid-build when the problem changes.

CORE COMPETENCIES

LLM + Agentic Systems

TypeScript + Node.js

Python + FastAPI

MCP + Multi-Model AI

REST APIs + Clean Architecture

Temporal.io + Kafka

AWS + GCP + Docker

0-to-1 Product Development

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Raleigh, NC

- Built multi-model LLM agent tooling across **Claude, GPT-4o, and Gemini** (8+ models, 3 providers) with MCP integration; implemented Temporal.io workflow orchestration and real-time event pipelines as core agentic system components
- Shipped 10+ backend features across **394 commits and 7 production deployments** -- TypeScript, Node.js, Express, PostgreSQL, Redis, Prisma, Auth0 (RBAC), Docker, Sentry, Playwright
- Built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 file storage) and composable REST APIs reviewed by and extending to non-engineer team contributors

NC State University Apr 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Designed a cloud-based genomic data platform (JBrowse 2) enabling 100+ researchers to interactively explore 500GB datasets; deployed on GCP Cloud Run
- Applied hyperparameter tuning and time-series analysis to ML models, improving accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows for healthcare claim automation using OCR, NLP, and ML; improved claim accuracy by 25% in a regulated domain with strict data handling requirements
- Automated cloud deployments on AWS and GCP; integrated Grafana monitoring dashboards

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led end-to-end SDLC for a data-driven support ticket system; distributed pipelines (Hadoop + Spark + Kafka) reduced latency by 20%

PROJECTS

AlgoPulse -- Agentic AI Orchestration Platform LLM + Temporal + Kafka github.com/Aditya-k24/AlgoPulse

Architected an event-driven agentic AI orchestration platform; identified API-polling bottleneck under load mid-build and pivoted to Supabase Edge Functions streaming AI reasoning loops directly to the client -- full 0-to-1 production rethink under ambiguity. Scaled worker architecture with Temporal.io and Kafka for state synchronization and fault tolerance, handling 2,000 async inference requests/second without state loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

Glance AI -- Compliance-Ready RAG System RAG + Regulated Domains github.com/Aditya-k24/GlanceAI

Built a compliance-ready RAG system for fintech and healthcare environments with per-tenant data isolation at the vector retrieval layer, preventing context leakage and LLM hallucinations. Designed for factual correctness in regulated data environments -- directly analogous to pharma's requirements for auditable, trustworthy AI outputs.

Python, LLMs, Vector Databases, RAG, per-tenant isolation

LLM Eval Harness -- Multi-Model Benchmark Claude + GPT-4o + Gemini github.com/Aditya-k24/llm-eval-harness

Built a vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 3 task types on a 2,000-example frozen test set. Metrics: ExactMatch, TokenF1, abstention accuracy, grounded hallucination rate, p50/p95 latency, cost/call; statistical significance via paired bootstrap CIs and McNemar test.

Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

VectorLift -- Semantic Search Engine Python + Kafka + FastAPI github.com/Aditya-k24/VectorLift

Production-grade semantic search on MS MARCO: Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant; FastAPI search server with per-stage latency tracking; evaluated 6 pipeline configurations (BM25, dense retrieval, cross-encoder re-ranking) with NDCG@10, MRR@10, MAP, and bootstrap significance testing.

Python, FastAPI, Kafka, Elasticsearch, Qdrant, PostgreSQL, Redis, Prometheus, Grafana, Docker

EDUCATION

North Carolina State University -- MS Computer Science Aug 2024 - May 2026

Raleigh, NC (Research Triangle) | Courses: Artificial Intelligence, Operating Systems, Software Engineering, Algorithms, Data Science

Mumbai University -- BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Structures & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

AI/LLM: Claude API, OpenAI, Gemini, MCP, LangChain, RAG, Temporal.io, Hugging Face, PyTorch

Languages: TypeScript, Python, Java, JavaScript, SQL **Backend:** Node.js, Express, FastAPI, REST APIs, Kafka, PostgreSQL, Redis, Docker

Cloud: AWS (S3, automation), GCP (Cloud Run), GitHub Actions CI/CD